CONAN: CONcept-oriented ANomaly Detector for Interpretable Zero-Shot Detection

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Abstract. Large vision-language models (LVLMs) have advanced zero-shot anomaly detection (ZSAD) by enabling generalization to unseen classes, yet most operate as semantic "black boxes" without explicit rationale. Incorporating concept bottleneck models (CBMs) is challenging because anomalies are condition-centric and transient, unlike stable object attributes. We propose **CONAN (CONcept-oriented ANomaly Detector)**, an interpretable framework that performs active concept curation through an object-agnostic concept bank organized into five macro-level anomaly dimensions. By aligning CLIP image features with category prototypes using LVLM-generated pseudo-labels, we inject fine-grained anomaly semantics into the embedding space. Experiments show strong zero-shot performance on industrial benchmarks, robust cross-domain generalization to medical datasets, and support for human-in-the-loop explanation refinement.

Keywords: Interpretable Zero-Shot Anomaly Detection · Concept Bottleneck Model · Human Intervention

1 Introduction

Anomaly detection plays a pivotal role in safety-critical domains like industrial inspection and medical diagnosis [31]. While recent methods have significantly improved detection performance, they predominantly operate as "black boxes," flagging anomalies without articulating the semantic rationale—specifically, *what* attributes constitute the defect. This lack of interpretability hinders practical deployment, where identifying the root cause is essential for accountability and targeted intervention [26]. Consequently, there is a pressing need for zero-shot anomaly detectors that not only generalize well but also expose their decisions through human-understandable concepts.

Concept bottleneck models (CBMs) bridge performance and transparency using interpretable concepts [19], yet they rely on stable, object-centric attributes.

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Extending this to anomaly detection is non-trivial because defects are *condition-centric*, describing the *state* of an object rather than its category [4]. Unlike stable object identities, anomaly concepts exhibit high visual heterogeneity across instances. Furthermore, VLMs like CLIP [24] are optimized for categorical semantics and often fail to align fine-grained anomaly conditions with visual features [33]. Consequently, directly transplanting conventional CBMs yields weak concept–anomaly alignment and limited interpretability.

In this work, we move to *active anomaly concept curation*. Instead of relying on unconstrained outputs, we construct a concept bank that organizes heterogeneous patterns into five condition-centric dimensions: *geometry*, *color*, *texture*, *surface integrity*, and *foreign substances*. For each category, we generate textual concepts using an LLM, embed them via CLIP, and aggregate them into category-level prototypes. This design is object-agnostic, capturing defect semantics shared across domains to enable robust condition-centric reasoning distinct from object-specific classification.

We propose **CONAN**, a CLIP-based CBM designed for interpretable zero-shot anomaly detection. Our framework follows a three-stage pipeline: First, we construct an anomaly concept bank to derive category prototypes from textual descriptions. Second, we perform alignment learning using pseudo-labels, aligning CLIP features with these prototypes via a classification loss to explicitly inject anomaly semantics. Third, we achieve localization through prototype–feature similarity, jointly optimizing image-level classification and pixel-level segmentation. During inference, CONAN utilizes an LVLM to generate natural language explanations based on intermediate predictions. Crucially, the framework supports human-in-the-loop intervention, allowing users to correct predicted categories to refine explanations for faithful analysis. Extensive experiments across diverse industrial and medical benchmarks demonstrate that CONAN achieves competitive or superior detection performance, validating its robust generalization capability.

2 Related Work

Anomaly Detection Traditional methods rely on learning the normal data distribution within a specific domain, utilizing reconstruction errors [27] or feature embeddings such as PaDiM [11] and PatchCore [25]. While effective, these approaches require target-domain training data and failed to generalize to unseen classes. The advent of VLMs has shifted the paradigm toward zero-shot anomaly detection. WinCLIP [17] and AnomalyCLIP [36] leveraged the aligned embedding space of CLIP [24] to detect anomalies using text prompts describing normal and abnormal states. However, these methods primarily function as black-box feature extractors sensitive to prompt engineering [35], often lacking transparency in the semantic reasoning process.

Interpretability in Anomaly Detection Interpretability is crucial in safety domains to ensure accountability. Traditional post-hoc techniques such as Grad-CAM [28] highlighted suspicious regions but failed to explain the semantic ratio-

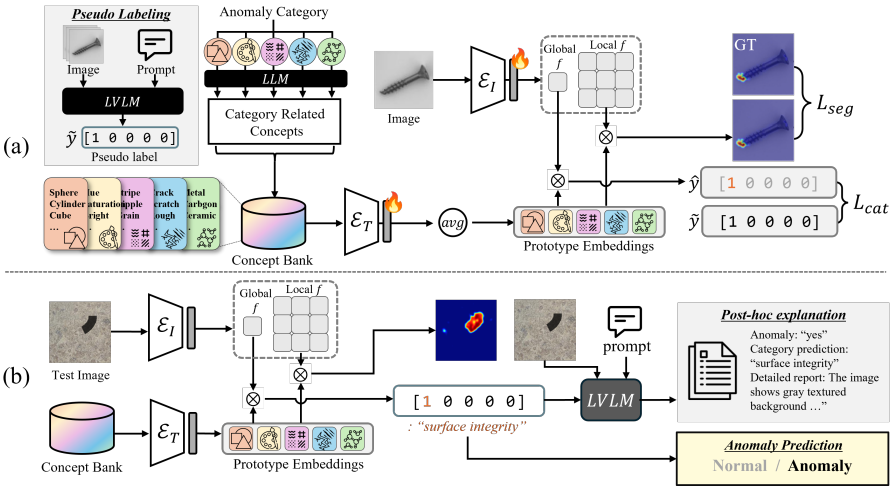

Fig. 1. Overview of the proposed framework CONAN: (a) Training and (b) Inference phase. We exclusively utilize the concept bank to conduct anomaly detection, with the concepts guided by an LLM, which requires no human annotations.

nale behind the decision [1]. Recently, LVLMs have been employed to generate natural language explanations for anomalies [13,34]. Although these approaches improve readability, the explanation generation is decoupled from the anomaly detection backbone, leading to low fidelity and a risk of hallucinations [18]. Unlike these methods, we propose an intrinsic interpretability framework where decision-making is directly grounded in defined anomaly concepts.

Concept Bottleneck Models CBMs offer a transparent learning paradigm by structuring predictions through human-interpretable concepts [19]. While successful in classification, extending these models to anomaly detection is non-trivial. Standard CBMs rely on stable *object-centric* attributes (e.g., has wings"), whereas anomalies require *condition-centric* concepts (e.g., "scratched," "contaminated") that describe state changes [4]. Furthermore, the embedding space of VLMs is often misaligned with such fine-grained conditions [33]. We address this by designing a concept bottleneck explicitly for anomaly understanding, organizing heterogeneous defects into structured semantic categories to enable transparent reasoning and user intervention.

3 Method

We introduce **CONAN**, a novel CBM designed to enable interpretable zero-shot anomaly detection. An overview of the proposed architecture is shown in Fig. 1.

3.1 Anomaly Category Concept Bank Construction

CBMs predict labels by combining activations of human-interpretable concepts from a predefined set, yet such concept banks are often dataset-specific and ex-

pensive to construct due to manual annotation. To mitigate this limitation, we adopt a label-free strategy that automatically generates concepts using an LLM. In addition, anomaly concepts differ markedly from the consistent, object-centric concepts found in conventional vision tasks [30]. Because anomalies manifest in diverse and context-dependent ways, narrowly defined, object-specific concepts are insufficient for robust reasoning. Accordingly, we focus on higher-level concepts that generalize across domains and object types, and introduce an anomaly-specific concept bank B_{ano_cat} that captures broad semantic patterns while accommodating substantial visual variability.

We categorize anomalies into five macro-level semantic groups: (1) *geo* for Geometry and shape deformation, (2) *color* for Color or photometric abnormalities, (3) *tex* for Texture irregularities and pattern distortions, (4) *surf* for Surface-level integrity damage (e.g., scratches, cracks, dents), and (5) *mat* for Material composition or foreign substance-related anomalies. The complete anomaly category set C_{ano} is defined as follows:

$$C_{ano} = \{C_{geo}, C_{color}, C_{tex}, C_{surf}, C_{mat}\}. \tag{1}$$

Each category contains a set of related textual concepts:

$$C_i = \{c_{i1}, c_{i2}, \ldots, c_{in_i}\}, \quad i \in \{\textit{geo, color, tex, surf, mat}\}. \tag{2}$$

Each concept c_{ij} is embedded using the CLIP text encoder $E_{\text{txt}}(\cdot)$, and the prototype representation for category C_i is computed as:

$$p_i = \frac{1}{n_i} \sum_{j=1}^{n_i} E_{\text{txt}}(c_{ij}), \tag{3}$$

where p_i denotes the centroid embedding of category C_i. We generate approximately 30 concepts per category using GPT-4o, following concept-size guidelines from prior research [29,30].

3.2 Category-Level Pseudo-Labeling and Alignment Learning

Unlike general domain settings, CLIP shows weak alignment between anomaly-related concept texts and anomaly images [23]. Therefore, we adopt supervised category alignment using pseudo-labels instead of relying on direct concept inference. We generated pseudo-labels using the LLaVA-1.5-7B [22] model within a prompt instructing the model to assign one of the predefined anomaly categories. Providing this contextual information enables the model to more accurately infer the anomaly type.

Based on these pseudo-labels, we conduct alignment learning. Let $f = E_{\text{img}}(X)$ denote the image embedding extracted by the CLIP image encoder. The aligned anomaly category is predicted by maximizing cosine similarity:

$$\hat{y} = \arg\max_i \cos(f, p_i), \quad \cos(f, p_i) = \frac{f \cdot p_i}{\|f\| \|p_i\|}. \tag{4}$$

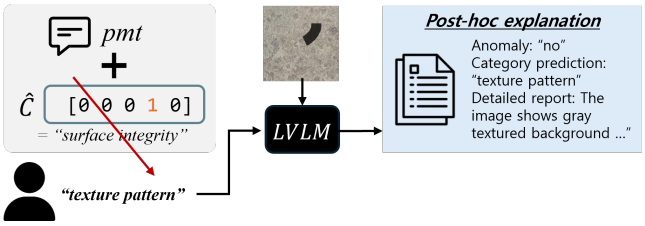

Fig. 2. Post-hoc user explanations for incorrectly predicted samples.

Therefore, the classification loss is computed using cross-entropy:

$$L_{\text{cat}} = -\sum_{i=1}^{N_c} y_i \log \sigma(\cos(f, p_i)). \tag{5}$$

3.3 Anomaly Localization via Prototype-Feature Similarity

Anomaly localization is performed by measuring the cosine similarity between the prototype embedding $p_{\hat{y}}$ and local image feature maps F_{local}:

$$H = \cos(F_{\text{local}}, p_{\hat{y}}), \tag{6}$$

where H denotes the anomaly heatmap. The localization loss is a combination of Dice loss [20] and Focal loss [21],

$$L_{\text{seg}} = L_{\text{Dice}}(H, M) + L_{\text{Focal}}(H, M). \tag{7}$$

where M represents the ground-truth mask. Therefore, the overall training objective is as follows:

$$L_{\text{total}} = L_{\text{cat}} + \lambda_{\text{seg}} L_{\text{seg}}. \tag{8}$$

3.4 Post-Hoc Explanation with User Intervention

During inference, our model generates: (1) the predicted anomaly category, (2) binary anomaly/normal decision, and (3) a localization heatmap. While the category prediction offers semantic interpretability, we further introduce a post-hoc explanation generation step using an LVLM. The LVLM takes the triplet $(X, \hat{C}, pmt)$, where $\hat{C}$ is the predicted category, and pmt is an instruction prompt, producing a natural-language explanation grounded in the model's anomaly reasoning.

To improve robustness and enable human-AI collaboration, we incorporate an optional user intervention mechanism (Fig. 2). If the predicted category is incorrect, the user may manually correct it, enabling the LVLM to regenerate a reliable explanation guided by the accurate category. This mechanism reinforces trust and flexibility of the overall system.

Table 1. Performance on in-domain datasets. Metrics grouped by Image-level (AUROC, AP%) and Pixel-level (AUROC, AUPRO%).

Task	*Image-level*				*Pixel-level*			
Method	VisA	MPDD	DAGM	DTD-Syn	VisA	MPDD	DAGM	DTD-Syn
CLIP-AC [9]	(65.0, 70.1)	(56.2, 66.0)	(82.5, 63.7)	(66.8, 83.2)	(47.8, 17.3)	(58.7, 29.1)	(32.7, 4.8)	(23.7, 5.5)
WinCLIP [17]	(78.1, 81.2)	(63.6, 69.9)	(91.8, 79.5)	(93.2, 92.6)	(79.6, 56.8)	(76.4, 48.9)	(87.6, 65.7)	(83.9, 57.8)
VAND [8]	(78.0, 81.4)	(73.0, 80.2)	(94.4, 83.8)	(86.4, 95.0)	(94.2, 86.8)	(94.1, 83.2)	(82.4, 66.2)	(95.3, 86.9)
AnomalyCLIP [36]	**(82.1, 85.4)**	(77.0, 82.0)	**(97.5, 92.3)**	(93.5, 97.0)	**(95.5, 87.0)**	(96.5, 88.7)	**(95.6, 91.0)**	**(97.9, 92.3)**
CONAN (Ours)	(79.6, 82.8)	**(77.6, 83.2)**	(96.5, 90.6)	**(94.8, 97.9)**	(94.4, 84.9)	**(96.6, 88.4)**	(95.0, 89.1)	(97.1, 90.2)

Table 2. Class-wise comparison on MVtec-AD2.

	Image-level		*Pixel-level*	
Class	AnomalyCLIP [36]	**CONAN** (Ours)	AnomalyCLIP [36]	**CONAN** (Ours)
can	(54.0, 56.6)	(56.1, 57.2)	(71.2, 25.8)	(54.1, 20.1)
fabric	(72.9, 80.6)	(77.2, 83.2)	(72.4, 36.7)	(72.8, 35.9)
fruit_jelly	(67.7, 88.5)	(62.8, 86.9)	(84.8, 58.3)	(85.5, 59.9)
rice	(72.6, 87.1)	(75.3, 88.2)	(89.3, 55.1)	(88.1, 46.6)
sheet_metal	(53.9, 84.7)	(60.7, 87.3)	(74.9, 9.5)	(68.7, 27.7)
vial	(52.1, 82.1)	(73.1, 91.3)	(88.4, 72.8)	(89.8, 71.5)
wallplugs	(41.1, 52.9)	(38.5, 52.7)	(83.0, 29.3)	(85.3, 41.7)
Mean	(59.2, 76.1)	**(63.4, 78.1)**	(**80.6**, 41.1)	(77.8, **43.4**)

4 Experiments

4.1 Experimental Setup

Datasets and Baselines We conduct experiments on a broad collection of industrial and medical anomaly detection benchmarks. Following prior studies, we use the test split of MVTec AD [5]—an industrial anomaly dataset—as the source data for training. When evaluating on this set, we instead use the VisA test split as the training source to avoid overlap.

We first evaluate on in-domain industrial datasets: VisA [37], MPDD [6], DAGM [32], DTD-Synthetic [3] and MVTec2 [15]. These datasets encompass manufactured objects and surface defects, spanning variations in texture, noise, and anomaly appearance. We further evaluate on out-of-domain medical datasets: HeadCT, BrainMRI [7], and Br35H [14] are used for image-level classification, while ISIC [10], Kvasir [2], Endo [16], and TN3K [12] provide pixel-level annotations for segmentation. This setting enables evaluation across different anomaly types, including lesions, tissue irregularities, and organ-level boundaries. We employed four state-of-the-art CLIP-based anomaly detection baselines for our experiments.

Implementation Details We follow the AnomalyCLIP configuration, initializing both the image and text encoders with the pretrained CLIP ViT-L/14 backbone. To preserve CLIP's prior knowledge while adapting to anomaly detection, we apply LoRA to both encoders and train only the low-rank adapters and newly added layers. Input images are resized to 518×518, and the model is trained with Adam using a batch size of 8 and a fixed learning rate of 1×10^{-3}. The segmentation loss weight λ_{seg} in Eq. (8) is set to 0.5. All models are trained

Table 3. Performance comparison on out-of-domain datasets. Metrics grouped by Image-level (AUROC, AP%) and Pixel-level (AUROC, AUPRO%).

Task	***Image-level***			***Pixel-level***			
Method	HeadCT	BrainMRI	Br35H	ISIC	Kvasir	Endo	TN3K
CLIP-AC [9]	(60.0, 60.7)	(80.6, 86.4)	(82.7, 81.3)	(36.0, 7.7)	(45.0, 16.8)	(46.6, 12.6)	(35.6, 5.2)
WinCLIP [17]	(81.8, 80.2)	(86.6, 91.5)	(80.5, 82.2)	(83.3, 55.1)	(69.7, 24.5)	(68.2, 28.3)	(70.7, 39.8)
VAND [8]	(89.1, 89.4)	(89.3, 90.9)	(93.1, 92.9)	(89.4, 77.2)	(75.0, 36.2)	(81.9, 54.9)	(73.6, 37.8)
AnomalyCLIP [36]	**(93.4, 91.6)**	(90.3, 92.2)	(94.6, 94.7)	(89.7, 78.4)	(78.9, 45.6)	(84.1, 63.6)	**(81.5, 50.4)**
CONAN (Ours)	(86.7, 84.9)	**(94.8, 96.1)**	**(95.7, 97.2)**	**(89.8, 80.3)**	**(82.1, 48.4)**	**(86.7, 84.9)**	(80.9, 48.6)

for 15 epochs running on four NVIDIA RTX 3090 GPUs (24GB). For evaluation metric, we use AUROC and AP for image-level anomaly detection, and AUROC and AUPRO for pixel-level localization.

4.2 ZSAD Performance Results

Industrial Domains Table 1 reports image- and pixel-level performance on four industrial benchmarks. Overall, CONAN shows competitive or superior results compared to existing CLIP-based methods.

For image-level classification, CONAN achieves the best scores on MPDD (77.6%, 83.2%) and DTD-Synthetic (94.8%, 97.9%), outperforming AnomalyCLIP on both datasets. On VisA and DAGM, CONAN is slightly behind AnomalyCLIP but clearly exceeds the remaining baselines. For pixel-level localization, CONAN maintains strong performance across all datasets. Although AnomalyCLIP records the highest overall scores, CONAN consistently outperforms the other baselines in both AUROC and AUPRO. These results indicate that category-level concept alignment supports robust recognition and localization while improving interpretability. Table 2 reports both image-level and pixel-level performance on the MVTec2 dataset, which includes particularly challenging classes due to images captured under diverse lighting conditions and a distribution that differs substantially from the original MVTec dataset. Despite these challenges, CONAN achieves an average improvement of 4.2% and 2.0% at the image level. At the pixel level, it shows a minor AUROC decrease of 2.4% on some classes but a 2.3% improvement on others. Overall, CONAN delivers performance that remains comparable to state-of-the-art black-box approaches, even under this difficult evaluation setting.

Medical Domains Table 3 summarizes image-level and pixel-level results on seven medical benchmarks. CONAN demonstrates strong cross-domain generalization and remains competitive or superior to prior CLIP-based approaches.

For image-level classification, CONAN surpasses all baselines on BrainMRI and Br35H, with gains of 4.5%p (AUROC) and 3.9%p (AP) over AnomalyCLIP on BrainMRI. On HeadCT, it is slightly weaker than AnomalyCLIP but still clearly outperforms the remaining methods. For pixel-level localization, CONAN performs well across ISIC, Kvasir, Endo, and TN3K, and exceeds AnomalyCLIP on several datasets—for instance, by 3.2%p (AUROC) and 2.8%p (AUPRO)

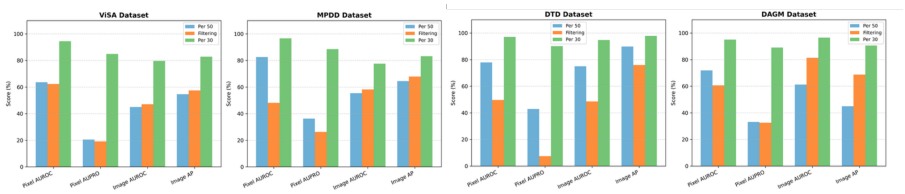

Fig. 3. Performance comparison by concept bank generation strategies.

Table 4. Evaluation of LVLM anomaly explanations with category guidance.

Dataset	Response	Acc.	Log.	Spec.	Rel.	Cat.	**Avg.**
VisA	LLaVA-1.5-7B [22]	3.4	5.0	4.8	5.2	2.8	4.2
	CONAN (Ours)	3.8	5.4	4.8	5.6	3.4	**4.6**
MPDD	LLaVA-1.5-7B [22]	6.0	6.8	6.0	7.2	5.8	**6.4**
	CONAN (Ours)	3.8	5.4	4.6	5.6	3.2	4.5
DAGM	LLaVA-1.5-7B [22]	5.8	6.6	5.8	6.4	5.2	6.0
	CONAN (Ours)	5.8	6.8	5.6	7.2	6.2	**6.3**
DTD	LLaVA-1.5-7B [22]	4.4	5.6	4.8	5.6	3.8	4.8
	CONAN (Ours)	4.6	6.4	5.4	6.4	4.2	**5.4**

on Kvasir. These observations confirm that category-level alignment transfers effectively to medical domains by focusing on condition-centric anomaly characteristics rather than object identity.

4.3 Analysis on Different Concept Bank Designs

Figure 3 compares three strategies for constructing the anomaly concept bank on industrial benchmarks: (1) generating 50 concepts per category and directly averaging them into prototypes; (2) generating 50 concepts and filtering them by a cosine similarity threshold around the prototype; and (3) generating 30 concepts per category without additional filtering (our default).

Using 30 concepts per category provides the strongest performance, while adding too many concepts reduces category separability and harms learning. Filtering performs worst, likely because it removes essential semantic coverage. Overall, effective anomaly modeling depends on maintaining a balanced, sufficiently diverse concept set per category.

4.4 Evaluation on Post-hoc Explanations

CONAN outputs anomaly scores, masks, and an explicit anomaly category, which can serve as a semantic anchor for post-hoc explanations. To assess the benefit of providing this category information, we compare explanations generated by LLaVA-1.5-7B in two settings: (1) using only the image, and (2) using both the image and the category predicted by CONAN. For each dataset, five images are randomly sampled and their explanations are evaluated by GPT-4o

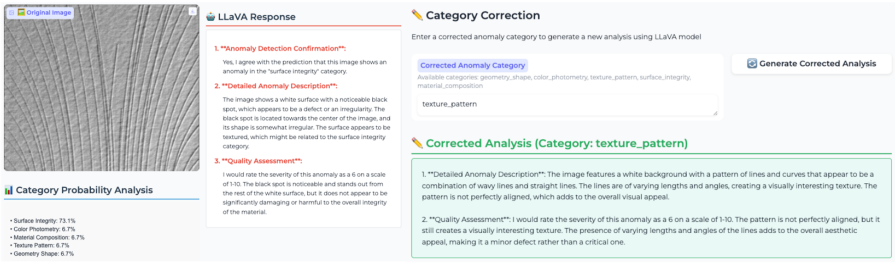

Fig. 4. User intervention example using gradio.

along five criteria: Accuracy (Acc.), Logical Coherence (Log.), Specificity (Spec.), Relevance (Rel.), and Category Classification Accuracy (Cat.). Table 4 summarizes the results for industrial datasets.

In three of the four datasets, providing CONAN's predicted category leads to higher overall explanation quality, indicating that the category acts as a helpful guidance signal for the LVLM. However, on MPDD, incorrect category predictions reduce explanation quality, emphasizing the need for optional user intervention when the predicted category is unreliable.

4.5 Case Study of User Intervention

We present a qualitative case study illustrating how user intervention can correct error cases and improve explanation quality. Figure 4 shows an example from an industrial dataset where the model initially predicts the category *surface integrity* for an image with subtle scratch-like line patterns on a cement surface. Using this predicted category, LLaVA generated an explanation that incorrectly describes a prominent "black spot" in the center of the image, which does not actually exist. By manually correcting the anomaly category from surface integrity to texture pattern, a human user enables the regenerated explanation to accurately focus on the irregular line structures, resulting in a more faithful and logically consistent description. This demonstrates that adjusting intermediate semantic representations provides an effective safeguard against error propagation and enhances the reliability of human–model interaction.

5 Conclusion

In this work, we addressed the opacity of zero-shot anomaly detection by introducing CONAN, a concept-oriented anomaly detection framework built around condition-centric anomaly concepts. We organized heterogeneous defects into a five-category concept bank, aligns CLIP image features with category prototypes using LVLM-generated pseudo-labels, and performs prototype-based localization under a joint classification–segmentation objective. Experiments on industrial and medical benchmarks showed that CONAN achieves competitive image and

pixel-level performance compared to prior CLIP-based methods, while preserving cross-domain generalization. By exposing explicit anomaly categories and guiding LVLM-based explanations through these concepts, our framework provides intrinsic interpretability and supports user intervention to correct category errors, improving explanation fidelity. We plan to conduct future work for refining the anomaly taxonomy, mitigating pseudo-label noise, and extending concept-centric anomaly reasoning to broader multi-modal settings. In future work, we plan to refine the anomaly taxonomy, mitigate pseudo-label noise, and extend concept-centric anomaly reasoning to broader multi-modal settings.

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References

1. Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M., Kim, B.: Sanity checks for saliency maps. LLaVA **31** (2018)
2. Ali, S., Jha, D., Ghatwary, N., Realdon, S., Cannizzaro, R., Salem, O.E., Lamarque, D., Daul, C., Riegler, M.A., Anonsen, K.V., Petlund, A., Halvorsen, P., Rittscher, J., de Lange, T., East, J.E.: Polyp dataset (2022), `https://figshare.com/articles/figure/Polyp_DataSet_zip/21221579`, figshare dataset, accessed: 2025-11-26
3. Aota, T., Tong, L.T.T., Okatani, T.: Zero-shot versus many-shot: Unsupervised texture anomaly detection. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. pp. 5564–5572 (2023)
4. Bergmann, P., Batzner, K., Fauser, M., Sattlegger, D., Steger, C.: The mvtec anomaly detection dataset: a comprehensive real-world dataset for unsupervised anomaly detection. International Journal of Computer Vision **129**(4), 1038–1059 (2021)
5. Bergmann, P., Fauser, M., Sattlegger, D., Steger, C.: Mvtec ad–a comprehensive real-world dataset for unsupervised anomaly detection. In: CVPR (2019)
6. Bergmann, P., Jin, X., Sattlegger, D., Steger, C.: The mvtec 3d-ad dataset for unsupervised 3d anomaly detection and localization. In: Proceedings of the 17th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2022) - Volume 5: VISAPP. pp. 202–213. INSTICC, SciTePress (2022). `https://doi.org/10.5220/0010865000003124`
7. Chakrabarty, N.: Brain mri images for brain tumor detection (2017), `https://www.kaggle.com/datasets/navoneel/brain-mri-images-for-brain-tumor-detection`, kaggle dataset, accessed: 2025-11-26
8. Chen, X., Han, Y., Zhang, J.: April-gan: A zero-/few-shot anomaly classification and segmentation method for cvpr 2023 vand workshop challenge tracks

1&2: 1st place on zero-shot ad and 4th place on few-shot ad. arXiv preprint arXiv:2305.17382 (2023)
9. Chen, X., Zhang, J., Tian, G., He, H., Zhang, W., Wang, Y., Wang, C., Liu, Y.: Clip-ad: A language-guided staged dual-path model for zero-shot anomaly detection. In: International Joint Conference on Artificial Intelligence. pp. 17–33. Springer (2024)
10. Codella, N., Rotemberg, V., Tschandl, P., Celebi, M.E., Dusza, S., Gutman, D., Helba, B., Kalloo, A., Liopyris, K., Marchetti, M., et al.: Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (isic). arXiv preprint arXiv:1902.03368 (2019)
11. Defard, T., Setkov, A., Loesch, A., Audigier, R.: Padim: a patch distribution modeling framework for anomaly detection and localization. In: International conference on pattern recognition. pp. 475–489. Springer (2021)
12. Gong, H., Chen, J., Chen, G., Li, H., Chen, F., Li, G.: Tn3k: Thyroid nodule ultrasound image dataset (2022), `https://github.com/haifangong/TRFE-Net-for-thyroid-nodule-segmentation`, open-access dataset (3493 ultrasound images), accessed: 2025-11-26
13. Gu, Z., Zhu, B., Zhu, G., Chen, Y., Tang, M., Wang, J.: Anomalygpt: Detecting industrial anomalies using large vision-language models. In: Proceedings of the AAAI conference on artificial intelligence. vol. 38, pp. 1932–1940 (2024)
14. Hamada, A.: Br35h: Brain tumor detection (2020), `https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection`, kaggle dataset, accessed: 2025-11-26
15. Heckler-Kram, L., Neudeck, J.H., Scheler, U., König, R., Steger, C.: The mvtec ad 2 dataset: Advanced scenarios for unsupervised anomaly detection. arXiv preprint arXiv:2503.21622 (2025)
16. Hicks, S.A., Jha, D., Thambawita, V., Halvorsen, P., Hammer, H.L., Riegler, M.A.: The endotect 2020 challenge: evaluation and comparison of classification, segmentation and inference time for endoscopy. In: International Conference on Pattern Recognition. pp. 263–274. Springer (2021)
17. Jeong, J., Zou, Y., Kim, T., Zhang, D., Ravichandran, A., Dabeer, O.: Winclip: Zero-/few-shot anomaly classification and segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 19606–19616 (2023)
18. Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y.J., Madotto, A., Fung, P.: Survey of hallucination in natural language generation. ACM computing surveys **55**(12), 1–38 (2023)
19. Koh, P.W., Nguyen, T., Tang, Y.S., Mussmann, S., Pierson, E., Kim, B., Liang, P.: Concept bottleneck models. In: International conference on machine learning. pp. 5338–5348. PMLR (2020)
20. Li, X., Sun, X., Meng, Y., Liang, J., Wu, F., Li, J.: Dice loss for data-imbalanced nlp tasks. In: Proceedings of the 58th annual meeting of the association for computational linguistics. pp. 465–476 (2020)
21. Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision. pp. 2980–2988 (2017)
22. Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. Advances in Neural Information Processing Systems **36**, 34892–34916 (2023)
23. Ma, W., Zhang, X., Yao, Q., Tang, F., Wu, C., Li, Y., Yan, R., Jiang, Z., Zhou, S.K.: Aa-clip: Enhancing zero-shot anomaly detection via anomaly-aware clip. In:

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 4744–4754 (2025)
24. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al.: Learning transferable visual models from natural language supervision. In: International conference on machine learning. pp. 8748–8763. PMLR (2021)
25. Roth, K., Pemula, L., Zepeda, J., Schölkopf, B., Brox, T., Gehler, P.: Towards total recall in industrial anomaly detection. In: Proceedings of the IEEE/CVF conference on Computer Vision and Pattern Recognition. pp. 14318–14328 (2022)
26. Ruff, L., Kauffmann, J.R., Vandermeulen, R.A., Montavon, G., Samek, W., Kloft, M., Dietterich, T.G., Müller, K.R.: A unifying review of deep and shallow anomaly detection. Proceedings of the IEEE **109**(5), 756–795 (2021)
27. Sakurada, M., Yairi, T.: Anomaly detection using autoencoders with nonlinear dimensionality reduction. In: Proceedings of the MLSDA 2014 2nd workshop on machine learning for sensory data analysis. pp. 4–11 (2014)
28. Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-cam: Visual explanations from deep networks via gradient-based localization. In: Proceedings of the IEEE International Conference on Computer Vision (Oct 2017)
29. Shang, C., Zhou, S., Zhang, H., Ni, X., Yang, Y., Wang, Y.: Incremental residual concept bottleneck models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 11030–11040 (2024)
30. Srivastava, D., Yan, G., Weng, L.: Vlg-cbm: Training concept bottleneck models with vision-language guidance. Advances in Neural Information Processing Systems **37**, 79057–79094 (2024)
31. Tao, X., Gong, X., Zhang, X., Yan, S., Adak, C.: Deep learning for unsupervised anomaly localization in industrial images: A survey. IEEE Transactions on Instrumentation and Measurement **71**, 1–21 (2022)
32. Wieler, M., Hahn, T., Hamprecht, F.A.: Weakly supervised learning for industrial opticalinspection (Sep 2007). `https://doi.org/10.5281/zenodo.12750201`, `https://doi.org/10.5281/zenodo.12750201`
33. Xie, C., Wang, B., Kong, F., Li, J., Liang, D., Zhang, G., Leng, D., Yin, Y.: Fg-clip: Fine-grained visual and textual alignment. In: International Conference on Machine Learning (2025)
34. Xu, J., Lo, S.Y., Safaei, B., Patel, V.M., Dwivedi, I.: Towards zero-shot anomaly detection and reasoning with multimodal large language models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 20370–20382 (2025)
35. Zhou, K., Yang, J., Loy, C.C., Liu, Z.: Learning to prompt for vision-language models. International Journal of Computer Vision **130**(9), 2337–2348 (2022)
36. Zhou, Q., Pang, G., Tian, Y., He, S., Chen, J.: Anomalyclip: Object-agnostic prompt learning for zero-shot anomaly detection. arXiv preprint arXiv:2310.18961 (2023)
37. Zou, Y., Jeong, J., Pemula, L., Zhang, D., Dabeer, O.: Spot-the-difference self-supervised pre-training for anomaly detection and segmentation. In: European Conference on Computer Vision. pp. 392–408. Springer (2022)